- **Current Status:** While large-scale quantum computers are still in development, early machines can already tackle specific, complex tasks that are impractical for classical computers.

4.4.2 Applications of Quantum Computing (In Brief)

- **Cryptography:** Quantum computers have the potential to break widely used cryptographic codes by factoring large numbers quickly-a task beyond the reach of current classical machines.
- **Drug Discovery and Chemistry:** They simulate molecular interactions at the quantum level, accelerating drug design and chemical research.
- **Optimization:** Industries use quantum algorithms to rapidly solve complex optimization problems in logistics, finance, and manufacturing.
- **Artificial Intelligence:** Quantum computing is expected to enhance machine learning by analyzing large, multidimensional data sets much faster.
- **Material Science and Physics:** Quantum simulations help researchers understand atomic-scale properties, fostering breakthroughs in new materials and energy systems.
- Quantum computing represents a paradigm shift, unlocking solutions to problems previously deemed unsolvable in practical timeframes, with widespread potential impact across science, industry, and security.

Multiple Choice Question With Answer

- **Choose the correct alternative from the following :**

(A) Immersive Technology – Concepts and Types

1. Immersive technology merges .....
   (a) Hardware and software
   (b) Physical and digital environments
   (c) Data and storage
   (d) Computers and networks
2. The main purpose of immersive technology is to .....
   (a) Replace physical systems
   (b) Enhance user experience through engagement
   (c) Reduce computer speed
   (d) Store more data
3. Which technology completely replaces the real world?
   (a) AR (b) MR
   (c) VR (d) XR
4. The device essential for Virtual Reality experience is .....
   (a) Monitor (b) VR headset
   (c) Smartphone (d) Keyboard
5. Augmented Reality works by .....
   (a) Simulating only virtual worlds
   (b) Overlaying digital objects on the real world
   (c) Creating sound-based simulations
   (d) Replacing human senses
6. Mixed Reality (MR) is a combination of .....
   (a) AR and AI (b) VR and IoT
   (c) AR and VR (d) None of these
7. Extended Reality (XR) includes .....
   (a) Only AR and VR (b) AR, VR, and MR
   (c) Only MR (d) None of these
8. Haptic technology mainly stimulates which human sense?
   (a) Hearing (b) Touch
   (c) Vision (d) Smell
9. A common example of AR in daily life is .....
   (a) Google Docs (b) Pokémon GO
   (c) YouTube (d) PowerPoint
10. Which immersive technology allows users to interact with both real and digital objects in real time?
    (a) AR (b) MR
    (c) VR (d) XR
11. The sense of "presence" in immersive technology means .....
    (a) The user is physically moving
    (b) The user feels part of the virtual world
    (c) The user's device is on
    (d) None of these
12. Which of the following is a wearable haptic device?
    (a) VR headset (b) VR gloves
    (c) Keyboard (d) Joystick
13. Touchable haptics are commonly used in .....
    (a) Smartphones and tablets
    (b) Servers
    (c) VR helmets
    (d) Data centers
14. Mid-air haptics deliver sensations using .....
    (a) Electricity (b) Ultrasound waves
    (c) Heat sensors (d) Vibrations only

15. Which immersive technology is most used for training pilots?
    (a) AR (b) VR
    (c) MR (d) XR
16. Which technology provides 3D visual overlays on physical spaces?
    (a) MR (b) AR
    (c) VR (d) Haptic
17. XR is best described as .....
    (a) Software simulation
    (b) The umbrella term for AR, VR, MR
    (c) Quantum device
    (d) Mechanical design tool
18. Which industry was among the first to adopt immersive technology widely?
    (a) Education (b) Gaming
    (c) Manufacturing (d) Agriculture
19. Immersive technologies engage which senses?
    (a) Sight and sound (b) Taste and smell
    (c) Balance and touch (d) None
20. Microsoft HoloLens is an example of .....
    (a) VR device (b) MR device
    (c) AR device (d) XR headset

(B) Applications of Immersive Technology

21. In education, immersive technology helps by .....
    (a) Creating physical textbooks
    (b) Enabling virtual field trips
    (c) Replacing teachers
    (d) Reducing school hours
22. Medical students use VR to .....
    (a) Play games
    (b) Practice surgery in virtual environments
    (c) Write reports
    (d) Scan patient IDs
23. AR can assist engineers by .....
    (a) Reducing machine cost
    (b) Overlaying design elements on real components
    (c) Replacing hardware
    (d) Printing 3D models
24. In real estate, VR enables .....
    (a) Cost estimation
    (b) Remote property tours
    (c) Document management
    (d) Security scanning
25. AR marketing helps customers by .....
    (a) Displaying product manuals
    (b) Trying products virtually
    (c) Increasing advertisements
    (d) None
26. In tourism, immersive technology offers .....
    (a) Virtual previews of destinations
    (b) Flight discounts
    (c) Paper tickets
    (d) Booking apps only
27. The military uses VR and AR to .....
    (a) Entertain troops
    (b) Conduct safe training simulations
    (c) Build bases
    (d) None
28. Healthcare uses VR for .....
    (a) Music therapy
    (b) Pain management and mental health
    (c) Billing
    (d) Insurance claims
29. In architecture, immersive technology allows .....
    (a) Viewing 3D models before construction
    (b) Enhancing paper drawings
    (c) Speeding up cement drying
    (d) Painting walls
30. AR glasses are used for .....
    (a) Showing digital info in real-world view
    (b) Listening to audio
    (c) Printing photos
    (d) Transferring files

(C) Green Computing

31. Green computing promotes .....
    (a) Energy conservation (b) Faster processors
    (c) Bigger servers (d) More power usage
32. Green IT mainly focuses on .....
    (a) Recycling and reducing e-waste
    (b) Producing expensive chips
    (c) Cloud gaming
    (d) Cryptocurrency mining
33. Energy-efficient hardware helps by .....
    (a) Increasing electricity cost
    (b) Reducing power consumption
    (c) Producing more heat
    (d) Degrading performance

34. Which is NOT a feature of green computing?
    (a) Energy saving (b) Reduced pollution
    (c) High carbon footprint (d) Recycling
35. Data centers improve cooling efficiency using .....
    (a) Hot aisle / cold aisle layout
    (b) Random arrangement
    (c) Closed chambers
    (d) Heating ducts
36. Liquid cooling in data centers is used for .....
    (a) Better heat transfer (b) Soundproofing
    (c) Data encryption (d) Visual enhancement
37. Virtualization helps green computing by .....
    (a) Reducing physical servers
    (b) Increasing storage devices
    (c) Running fewer applications
    (d) Wasting resources
38. Green computing is also called .....
    (a) Cloud IT (b) Green IT
    (c) Smart IT (d) Energy IT
39. Which of these is a benefit of green computing?
    (a) Cost savings (b) High temperature
    (c) More pollution (d) Data loss
40. A paperless office contributes to .....
    (a) Waste generation (b) Green computing
    (c) Carbon increase (d) None

(D) E-Waste Management and Recycling :

41. The first stage in e-waste management is .....
    (a) Sorting (b) Collection
    (c) Dismantling (d) Recycling
42. Data destruction ensures .....
    (a) File recovery (b) Information security
    (c) Power efficiency (d) Free software
43. Shredding is done to .....
    (a) Mix components
    (b) Break down parts for sorting
    (c) Repair devices
    (d) None
44. Magnetic separation removes .....
    (a) Plastic (b) Steel and iron
    (c) Copper (d) Glass
45. Eddy current separation targets .....
    (a) Non-ferrous metals (b) Glass
    (c) Wood (d) Paper
46. Water separation helps in separating .....
    (a) Glass and plastics (b) Metals
    (c) Chemicals (d) All materials
47. Material recovery extracts .....
    (a) Gold, copper, silver (b) Plastic, paper, wood
    (c) Lead, mercury (d) Glass only
48. Hazardous waste includes .....
    (a) Plastic
    (b) Lead, mercury, cadmium
    (c) Paper
    (d) Aluminum
49. The purpose of e-waste recycling is to .....
    (a) Produce more e-waste
    (b) Reduce environmental impact
    (c) Generate new waste
    (d) Discard all materials
50. Recovered materials are used to .....
    (a) Create new electronic products
    (b) Build buildings
    (c) Produce paper
    (d) Make fuel

(E) Quantum Computing :

51. Quantum computing is based on .....
    (a) Quantum mechanics (b) Classical physics
    (c) Thermodynamics (d) Nanotechnology
52. The basic unit of quantum information is .....
    (a) Bit (b) Byte
    (c) Qubit (d) Node
53. Superposition allows a qubit to be .....
    (a) 0 or 1 (b) 0 and 1 simultaneously
    (c) 2 only (d) None
54. Entanglement means .....
    (a) Two qubits interact instantly
    (b) Qubits separate
    (c) Bits collide
    (d) Data loss
55. Quantum interference helps .....
    (a) Eliminate wrong solutions
    (b) Store memory
    (c) Increase heat
    (d) Encrypt data
56. Classical computers use .....
    (a) Qubits (b) Bits
    (c) Gates only (d) Photons

57. Quantum computers can solve problems .....
(a) Sequentially (b) In parallel
(c) Randomly (d) Slowly

58. Example of a quantum computer .....
(a) IBM Quantum (b) Dell Laptop
(c) Raspberry Pi (d) Chromebook

59. Quantum computing helps in .....
(a) Drug discovery (b) Game design
(c) Web hosting (d) UI design

60. Quantum Key Distribution (QKD) is used in .....
(a) Cryptography (b) AI
(c) Robotics (d) Machine learning

61. Google's quantum computer is called .....
(a) Titan (b) Sycamore
(c) QuantumX (d) DeepMind

62. Quantum computers process information using .....
(a) Classical gates (b) Quantum gates
(c) Optical drives (d) CPUs

63. Quantum computing benefits which field most?
(a) Optimization (b) Animation
(c) Gaming (d) Cloud hosting

64. Quantum computing is still in .....
(a) Early development stage
(b) Full production
(c) Decline
(d) Decommissioning

65. A key difference between classical and quantum computing is .....
(a) Speed of data transfer
(b) Representation of data (bits vs qubits)
(c) Screen resolution
(d) Hardware design

66. In quantum computing, probability is used for .....
(a) Observing results (b) Compressing data
(c) Encrypting data (d) Deleting files

67. Quantum algorithms are faster for .....
(a) Sorting (b) Factoring large numbers
(c) Simple addition (d) Text processing

68. Which principle allows linked qubits to share information instantly?
(a) Superposition (b) Entanglement
(c) Holography (d) Refraction

69. Which area can quantum computing revolutionize?
(a) Cybersecurity (b) Photography
(c) Social media (d) Printing

70. The main challenge in building quantum computers is .....
(a) Data storage
(b) Qubit stability and error correction
(c) Color display
(d) User interface

Answers

1. (a)	2. (b)	3. (c)	4. (b)	5. (b)	6. (c)	7. (b)
8. (b)	9. (b)	10. (b)	11. (b)	12. (b)	13. (a)	14. (b)
15. (b)	16. (a)	17. (b)	18. (b)	19. (a)	20. (b)	21. (b)
22. (b)	23. (b)	24. (b)	25. (b)	26. (a)	27. (b)	28. (b)
29. (a)	30. (a)	31. (a)	32. (a)	33. (b)	34. (c)	35. (a)
36. (a)	37. (a)	38. (b)	39. (a)	40. (b)	41. (b)	42. (b)
43. (b)	44. (b)	45. (a)	46. (a)	47. (a)	48. (b)	49. (b)
50. (a)	51. (a)	52. (c)	53. (b)	54. (a)	55. (a)	56. (b)
57. (b)	58. (a)	59. (a)	60. (a)	61. (b)	62. (b)	63. (a)
64. (a)	65. (b)	66. (a)	67. (b)	68. (b)	69. (a)	70. (b)